

The Information Order of Experiments: Intervention Lattices and Generator Identifiability

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Abstract

Which experiment should you run next? We give the question a finite, exactly computable form. For a hypothesis space $\mathcal{H}$ of candidate generators, write $O_1 \preceq O_2$ when O_1 is obtainable from O_2 by garbling. Refinement then contracts preimages, $P_{O_2}(g) \subseteq P_{O_1}(g)$, so per-generator ambiguity $i_O(g)$ and its population mean $H(G | O)$ both fall—a one-line consequence, but one that turns “more informative” into an order rather than a scale.

We instantiate the order exactly on all 1,500,625 four-state partially observed Markov chains, enumerating every subset of hidden-state resets. The reset sets form a Boolean lattice under inclusion; informativeness itself is only a preorder, and we reserve “lattice” for the former. Four findings. The passive experiment is a *deterministic* coarsening of the full interventional one, confirmed exactly, so its 1.874 bits of shortfall is precisely $I(G; O_I | O_P)$. Informativeness is genuinely partial, not total: a single reset leaves 4.504 bits where passive observation leaves 3.245, so one intervention is *worse* than watching, while two are better. Which two matters enormously—two resets inside one observational lump identify 0% of the class, two that cross it identify 58.55%, at identical cost—though a sweep over all 15 partitions of the state space shows the crossing rule holds only when the observation’s blocks are equal in size, and reverses when they are not. And information gain is submodular across every triple we can form, which places greedy experiment selection on familiar footing.

The organising claim is that resolving a mechanism is partition refinement, that the order on experiments controls its rate, and that the interventions worth buying are the ones that cross distinctions the observation map erases.

1 Introduction

A companion paper [7] argues that identifiability should be measured rather than assumed, by counting the preimage $P_O(g) = \{h \in \mathcal{H} : O(h) = O(g)\}$ of an observation map, and reports that identifiability and predictive uncertainty are distinct coordinates. It leaves a question open. Identifiability there is relative to a declared observation map and a declared class of admissible interventions, and changing that class moved 82% of one hypothesis space between regimes. If knowability is regime-relative, the regimes themselves need a structure.

They have one. Experiments are partially ordered by informativeness, and the order is old: Blackwell’s comparison of experiments [1] ranks one experiment above another when the second can be obtained from the first by a stochastic garbling. What we add is a finite, exactly enumerable instance in which the whole lattice can be computed rather than argued about, together with three empirical facts that are not consequences of the order alone.

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2 The refinement order

Let $\mathcal{H}$ be finite, G uniform on $\mathcal{H}$, and let O_1, O_2 be observation maps.

Definition 1. $O_1 \preceq O_2$ iff there is a map F with $O_1 = F \circ O_2$.

Proposition 2 (Containment). *If $O_1 \preceq O_2$ then for every $g \in \mathcal{H}$, $P_{O_2}(g) \subseteq P_{O_1}(g)$, hence $i_{O_2}(g) \leq i_{O_1}(g)$ and $H(G | O_2) \leq H(G | O_1)$.*

Proof. If $O_2(h) = O_2(g)$ then $O_1(h) = F(O_2(h)) = F(O_2(g)) = O_1(g)$. Containment gives the inequality on $\log_2$ of cardinalities pointwise, and averaging preserves it. $\square$

Proposition 3 (Value of refinement). *If $O_1 \preceq O_2$ then $H(G | O_1) - H(G | O_2) = I(G; O_2 | O_1)$.*

Proposition 3 is what licenses reading a difference of conditional entropies as the information the finer experiment carries that the coarser one cannot reach. It requires the order; between incomparable experiments the difference is a comparison of two numbers and nothing more.

Proposition 4 (Submodularity). *Let $Z_j = O_j(G)$ for j in a finite index set J , with each Z_j a deterministic function of G , and write $Z_A = (Z_j)_{j \in A}$. Then $F(A) = I(G; Z_A) = H(Z_A)$ is monotone and submodular in A . The same holds conditionally on any fixed observation P : $F_P(A) = I(G; Z_A | P) = H(Z_A | P)$ is monotone and submodular.*

Proof. $H(Z_A | G) = 0$ because each Z_j is a function of G , so $I(G; Z_A) = H(Z_A) - H(Z_A | G) = H(Z_A)$. The entropy of a collection of random variables is a monotone submodular set function, which is Shannon's inequality $H(Z_{A \cup \{j\}}) - H(Z_A) \geq H(Z_{B \cup \{j\}}) - H(Z_B)$ for $A \subseteq B$. Conditioning on a fixed P preserves both properties, and $I(G; Z_A | P) = H(Z_A | P)$ by the same argument. $\square$

The pair (J, r) with $r(A) = H(Z_A)$ is therefore an *entropic polymatroid*, the dependence structure Fujishige identified for collections of random variables [2]. This is more specific than “some submodular objective”: $r(\{j\}) = H(Z_j)$ is the standalone information of experiment j , $r(A \cup \{j\}) - r(A) = H(Z_j | Z_A)$ its marginal information, and $r(\{i\}) + r(\{j\}) - r(\{i, j\}) = I(Z_i; Z_j)$ the pairwise redundancy. Our contribution is not the greedy machinery, which is standard in observation selection [4], but that the finite-generator intervention problem induces an explicit entropic polymatroid whose rank values are exactly computable from generator equivalence classes, making those guarantees directly applicable.

Proposition 5 (One-way migration). *Classify generators by thresholds on $i_O(g)$ and on a behavioural coordinate $p_k(g)$. If p_k is evaluated under an operating condition held fixed across $O_1 \preceq O_2$, then under refinement no generator can become less identifiable, and none can move in the behavioural coordinate.*

This is worth stating because it bounds what a migration statistic can mean. Reported migration under refinement is necessarily one-dimensional and one-directional; it is not evidence of two-dimensional mobility.

3 The benchmark system

We reuse the family of [7, §9] rather than introduce another: hidden state $X_t \in \{0, 1, 2, 3\}$, transition rows drawn from $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$ summing to one (35 rows), all $35^4 = 1,500,625$ generators enumerated completely, observation lumping $O(0) = O(1) = 0$, $O(2) = O(3) = 1$, horizon $T = 6$. The passive experiment starts from the uniform hidden state. For $A \subseteq \{0, 1, 2, 3\}$, the experiment O_A resets the hidden state to each $j \in A$ in turn and returns the resulting observable laws.

4 Results

The passive experiment is a deterministic coarsening. Writing L_j for the law obtained by resetting to state j , we verified exactly, over every generator, that

$$O_P = \frac{1}{4} \sum_j L_j,$$

so $O_P \preceq O_{\{0,1,2,3\}}$ with a deterministic F . Proposition 2 then applies, and we confirm its conclusion computationally as an implementation check: zero containment violations across every pair $A \subset B$, and zero against the passive mixture.

Reset set A	$H(G \mid O_A)$	classes	$\Pr[i_{O_A} = 0]$	$\max P $
$\emptyset$	20.517	1	0.00%	1,500,625
any singleton	4.504	400,584	0.00%	67,375
$\{0, 1\}$ or $\{2, 3\}$	3.356	482,902	0.00%	30,625
$\{0, 2\}, \{0, 3\}, \{1, 2\}, \{1, 3\}$	2.618	967,107	58.55%	8,953
any triple	1.914	1,101,899	69.84%	7,263
$\{0, 1, 2, 3\}$	1.371	1,235,809	82.01%	6,561
passive (mixture)	3.245	303,229	0.00%	7,537

Table 1: The intervention lattice, exact over 1,500,625 generators. Rows within a group are identical to the digits shown, reflecting the symmetry of the construction.

Informativeness is partial, not total. A single reset leaves 4.504 bits; passive observation leaves 3.245. One intervention is *less* identifying than watching a system you did not touch. There is no contradiction with Proposition 2: O_P is a coarsening of the full reset set, not of any singleton, so the two are incomparable in $\preceq$ and the proposition says nothing about them. Two resets (2.618) overtake passive; one does not. Three relations are in play and only two are comparabilities:

$$O_P \preceq O_{\{0,1,2,3\}}, \quad O_{\{0\}} \preceq O_{\{0,1,2,3\}}, \quad O_{\{0\}} \text{ and } O_P \text{ incomparable.}$$

Passive is a garbling of the full reset protocol, but neither passive nor a single reset is a garbling of the other, so their entropies may fall in either order and here they fall the surprising way. Any claim of the form “intervention beats observation” is false as stated; the correct statement is that informativeness is a partial order.

The interventions that pay are those crossing the observation’s kernel. Let $x \sim_O y$ iff $O(x) = O(y)$, here $\{0, 1\}$ and $\{2, 3\}$. Two resets chosen inside one $\sim_O$ class leave 3.356 bits and identify *nothing*; two that cross leave 2.618 bits and uniquely identify 58.55% of the entire hypothesis class. The cost is identical. Greedy selection, run without knowledge of the lumping, chooses 0, then 2, then 3, then 1—crossing at the first opportunity.

We stated this as a conjecture—interventions crossing $\sim_O$ dominate interventions of equal cardinality confined within one class—and then tested it, with criteria fixed in advance, over *all* 15 set partitions of the four hidden states. It is false.

The sweep uses a coarser but still completely enumerated family so that every partition is affordable: transition rows from $\{0, \frac{1}{2}, 1\}$, giving $10^4 = 10,000$ generators, at $T = 5$; entropies are therefore not comparable in absolute terms with Table 1. Within each (partition, $|A|$) cell we asked whether mean $H(G | O_A)$ is non-increasing in the number of blocks A meets. It is in only 3 of 23 cells—13%, against a pre-committed threshold of 90%.

The failures have a clean shape. For the partition $\{0\} | \{1, 2, 3\}$ the effect *reverses*: two resets confined to the large block leave 4.852 bits, while two that cross leave 5.917—crossing is worse by 1.065 bits. For the balanced $\{0, 1\} | \{2, 3\}$ the original direction holds, 3.799 against 4.187. Restricting the test to partitions whose blocks are all the same size, it passes 3/3.

No replacement conjecture is required, because the exact rule is already available. Since each Z_j is a deterministic function of G , Proposition 4 gives $F(A) = H(Z_A)$, so the value of adding reset j to a set A is exactly its conditional novelty

$$\Delta(j | A) = H(Z_j | Z_A),$$

and for a pair, $H(Z_i, Z_j) = H(Z_i) + H(Z_j) - I(Z_i; Z_j)$. A pair of interventions is valuable when both carry information and little of it is shared. Block crossing was only ever a proxy for low redundancy, and decomposing the two cases (Experiment 25) shows they are governed by different terms.

Partition	pair	crossing	$H(Z_i)+H(Z_j)$	$I(Z_i; Z_j)$	$H(G O_A)$
$\{0\} \{1, 2, 3\}$	$\{1, 2\}$	no	12.056	3.621	4.852
	$\{0, 1\}$	yes	11.045	3.675	5.917
$\{0, 1\} \{2, 3\}$	$\{0, 1\}$	no	14.342	5.241	4.187
	$\{0, 2\}$	yes	14.342	4.854	3.799

Table 2: The crossing effect and its reversal, decomposed by $H(Z_i, Z_j) = H(Z_i) + H(Z_j) - I(Z_i; Z_j)$.

On the balanced partition the individual entropies are *exactly equal*, so the entire 0.387-bit advantage of crossing is a redundancy effect: crossing pairs share less (4.854 against 5.241). On the unbalanced partition the redundancies are nearly equal—and the crossing pair is marginally the *more* redundant of the two—so essentially all of the 1.065-bit gap, 1.011 of it, comes from the individual term: a reset into a singleton observational block carries only 5.017 bits where a reset into the three-state block carries 6.028.

We had guessed the singleton reset was redundant. It is not; it is simply less informative on its own, which is a different mechanism and one the decomposition settles rather than argues. Crossing is a heuristic for low redundancy that works when symmetry equalises the individual terms and fails when it does not.

The selection rule that follows is exact rather than heuristic:

$$j^* = \arg \max_{j \notin A} H(Z_j | Z_A),$$

which is the greedy step of Proposition 4 written in the variables that make it computable. What remains genuinely open is not the rule but its explanation: *which structural features of $(\mathcal{H}, O)$ make $H(Z_j | Z_A)$ large?* Observation block size re-enters there as a candidate explanatory variable rather than as the theory. A natural refinement to study is the interventional signature partition $x \sim_Z y \iff Z_x = Z_y$, which is not the observational partition $\sim_O$: states alike under $\sim_O$ but distinct under $\sim_Z$ are exactly the distinctions intervention can unlock, and states alike under both are hidden from this experimental family altogether.

We record the refutation rather than quietly narrowing the claim, because the conjecture as first written was the natural reading of Table 1 and would have been an easy thing to leave standing.

Information gain is submodular—by Proposition 4, not by observation. An earlier version of this paper reported 108 submodularity inequalities holding with zero violations and described the property as measured. That was a category error. In this setting the observable law of each reset is an exact deterministic function of the generator, so $F(A) = H(Z_A)$ and submodularity is Shannon’s inequality; the computation could not have failed, and it checks our implementation rather than the mathematics. We record it as such: 108/108, consistent with the theorem.

What the theorem buys is operational. Maximising a monotone submodular function under a cardinality constraint admits the classical $(1 - 1/e)$ greedy guarantee [6], so under a budget $|A| \leq k$ the rule

$$j_t^* = \arg \max_{j \notin A_t} I(G; Z_j | P, Z_{A_t})$$

—perform next the experiment expected to eliminate the most remaining ambiguity—is within $1 - 1/e$ of the best possible set of that size. That holds for any family of experiments whose exact outputs are deterministic in the unknown generator, which is the regime this framework assumes throughout. The greedy gain sequence here is 16.013, 1.886, 0.704, 0.543 bits; the fourth reset is nearly redundant.

The polymatroid has no non-trivial closure. A polymatroid carries a closure operator, $j \in \text{cl}(A)$ iff $H(Z_j | Z_A) = 0$, and a basis is a minimal spanning set. We computed these exactly— $H(Z_J | Z_A) = 0$ holds iff A induces as many generator classes as J , an integer comparison rather than a tolerance—for all 14 non-trivial partitions of the four hidden states. In every one, the *only* spanning set is J itself. There is no experimental compression here: no reset is redundant given the other three, and a four-reset protocol cannot be replaced by a smaller one.

The margins by which this fails are informative. $H(Z_j | Z_{J \setminus \{j\}})$ ranges from 0.280 to 0.902 bits, and within every partition it is monotone in the size of j ’s observation block, with zero violations across all 14: a reset into a state the observation already isolates contributes least that the others cannot supply. Notably block size does *not* predict $H(Z_j)$ itself—Experiment 25 found those means non-monotone at 8.050, 8.157, 6.028 for block sizes 1, 2, 3—so the structural variable governs the conditional contribution and not the standalone one. This is the defensible residue of the intuition that failed in Experiment 24, and it is worth noting that the pairwise

redundancies did not reveal it: $I(Z_0; Z_1)$ was marginally *lower* for the singleton in the reversal case, while the conditional-on-all-others measure separates cleanly.

A further contrast: the rank values $r(J)$ vary widely with the observation partition, from 9.892 to 13.288 bits, while the basis structure does not vary at all. What the observation map changes is how much the experiments tell you, not which of them are needed.

Compression exists, but it is a property of the hypothesis class. The absence of compression above is suspicious, because $\mathcal{H} = \text{ROWS}^4$ is a product: the four transition rows are chosen independently, which is exactly the structure that should resist it. We therefore compared three classes on the same system and the same 14 partitions, with criteria fixed in advance. Alongside the product class we took a *tied* class constraining row 3 to equal row 0, giving 10^3 generators, and a *random* subset of the product class of the same cardinality. The random class is the control: it decides whether any compression is structural or merely an effect of having fewer generators to tell apart.

Hypothesis class	$ \mathcal{H} $	partitions compressing	minimal bases
product, rows free	10,000	0/14	$\{0, 1, 2, 3\}$
tied, row 3 = row 0	1,000	14/14	$\{0, 1, 2\}, \{1, 2, 3\}$
random subset, same size	1,000	0/14	$\{0, 1, 2, 3\}$

Table 3: Compression is structural, not a cardinality effect: the tied class compresses in every partition while a random class of identical size compresses in none.

The tied class compresses in every partition, always to a basis of size three, and always by omitting one of the two tied states—never states 1 or 2, whose rows remain free. The equally sized random class compresses nowhere. So Experiment 26’s result is a statement about the hypothesis class, not about interventions: *an intervention is redundant exactly when the generators carry a dependency that makes it so.*

Two dissociations follow, and together they say where each structure lives. Changing the observation map moves the rank values $r(A)$ but leaves the basis untouched; introducing a dependency among generators leaves the qualitative rank profile recognisable but collapses the basis. The experiment family has a small informational basis precisely when the generator family is constrained.

The value of full intervention. $H(G | O_P) - H(G | O_I) = 3.245 - 1.371 = 1.874$ bits, which by Proposition 3 is exactly $I(G; O_I | O_P)$: the generator information the complete reset protocol carries that no amount of passive observation can reach.

5 Related work

The order is Blackwell’s [1]; our instance is its degenerate deterministic case. Choosing experiments to maximise expected information is Lindley’s [5], and choosing interventions to secure identifiability is well developed in causal discovery, where optimal strategies are known [3]. The greedy guarantee for submodular objectives is [6]. What we contribute is not a new criterion but an exactly enumerated lattice: every experiment in a finite family, every containment relation

checked rather than assumed, and the resulting empirical facts—incomparability, the crossing rule, submodularity—measured rather than argued.

6 Limitations

The lattice is computed on one family with one lumping, and the striking symmetry of Table 1 is a property of that construction. The compression results of Table 3 use one dependency—a single tied row—and we have not characterised which dependencies produce which bases. The crossing conjecture is refuted in general and survives only for equal-block partitions; the exact selection rule $\Delta(j \mid A) = H(Z_j \mid Z_A)$ replaces it, but we have no structural characterisation of when that quantity is large, which is the open problem this paper leaves. Submodularity is a theorem here (Proposition 4) and holds whenever exact experiment outputs are deterministic functions of the generator; it would need revisiting for noisy or sampled observations, where $H(Z_A \mid G) > 0$ and the identity $F(A) = H(Z_A)$ fails. Identification is at horizon $T = 6$, which is where the partition of this family stabilises [7, §9], but stability over a finite window is evidence rather than proof. Proposition 2 is a theorem, so the computation confirming it tests our implementation, not the mathematics.

7 Discussion

Resolving a mechanism is partition refinement. Beginning with $[g] = \mathcal{H}$, each experiment induces an equivalence relation and the cell containing the truth contracts,

$$\mathcal{H} \supseteq [g]_{\Pi_1} \supseteq [g]_{\Pi_2} \supseteq \cdots,$$

with perfect resolution at $[g]_{\Pi} = \{g\}$ and irreducible ambiguity when the admissible order bottoms out while $|[g]_{\Pi}| > 1$. Three structures sit on top of one another: generator space is partitioned by an observation; the information order says which partitions refine which; and submodularity controls how efficiently refinements can be bought under a budget. What the measurements add is that the rate of contraction is controlled by the order on experiments, that the order is partial rather than total—so more control is not uniformly more information—and that the experiments worth buying are the ones that cross the distinctions the observation map erases.

Reproducibility. `mgs_lattice.py` produces the intervention-order figures and `mgs_partitions.py` the partition sweep of §4; `audit.py` re-derives every number here against them, and reports its status in `AUDIT.txt`. All are in the repository accompanying [7].

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